The Authority Transition Algebra

The Lifecycle of Machine Authority: States, Operators, Contraction Laws, and Temporal Guarantees

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The mathematical foundation for the Authority Handoff / Withdrawal / Recovery Protocol and the Evidence–Authority Coupling Layer: labeled transition semantics, an operator calculus over authority states, safety and liveness properties, recovery integrity, and the dynamical analysis of authority under evidence decay. Authority is not a permission flag but a state in a formal transition system — granted, contracted, suspended, revoked, and recovered only through named operators with discharged preconditions.

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C0 INFORMATIONAL · MATHEMATICAL SPECIFICATION — ALL PROPERTIES STATED AS PROOF OBLIGATIONS UNTIL DISCHARGED

DOCUMENT CONTROL & EPISTEMIC CONTRACT

0.1 Provenance and Position in the Program

This is Paper 1 of the research-and-build program defined by the *SVT Architect Analysis* (source document I): the Authority Algebra and State Machine, deepening Paper 0 (*Constitutional Intelligence: A Formal Specification of Bounded Machine Authority*, v1.0, 2026-10-03) from architectural specification to mathematical object. Paper 0 defined *what* the lifecycle is; this document defines what it *must provably do*. Its two stated burdens, per the source: to be the mathematical foundation of the Authority Handoff / Withdrawal / Recovery Protocol, and of the Evidence–Authority Coupling Layer. It reconciles against the *SAVANT PRIME Capability Inventory v1.0* (source document II) under the same rule as Paper 0 — where documents disagree in emphasis, adopt the stricter constraint and record the adoption (§11).

0.2 Epistemic Contract

Paper 0's four statuses carry over unchanged, with one sharpening for this document's genre: every property asserted of the algebra is a **proof obligation**, not a theorem. A property named "T-x" in this document is a claim that a proof or model-checking verdict *should exist* — accompanied by the exact conditions under which that expectation fails. Until the reference implementation's model checker discharges them, all such properties are CONJECTURE, however plausible their sketches read. The tags:

TAG	MEANING
VERIFIED-FROM-SOURCE	Traceable to the two source artifacts or to Paper 1, by section.
FORMALIZED-HEREIN	New formal structure authored here. Checkable; unvalidated.
CONJECTURE	Property with a stated proof/model-check obligation and falsification condition.
UNKNOWN	Open question, represented explicitly.

0.3 Authority Boundary

C0 informational. This document may be used to implement, verify, and attack the algebra. It may not be used as evidence that any system built on the algebra is correct; that evidence is precisely what the proof obligations of §4–§7 exist to produce, and they are undischarged at seal time.

0.4 Falsification Posture

The algebra is designed so that its most important properties are *decidable*: the lifecycle automaton is finite, the operator preconditions are checkable predicates, and every safety property of §5 is a reachability statement a model checker can kill. If the machine survives, the

theory stands; if the machine finds one counterexample, the theory changes. That asymmetry is the document's honesty mechanism.

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PART I — THE MACHINE

1 From Specification to Algebra

Chunk 1 specified the kernel as architecture: states, operators, invariants, gating rules. Architecture tells an engineer what to build. It does not tell anyone — engineer, auditor, adversary, or regulator — what the built thing *guarantees*. Guarantees require a mathematical object whose properties can be proven or refuted. This document constructs that object: the authority lifecycle as a labeled transition system with denotational operator semantics, and states the properties that make it worthy of the word "constitutional." FORMALIZED-HEREIN

1.1 Why a second document

The source assigns Paper 1 two foundations to lay VERIFIED-FROM-SOURCE: the Authority Handoff / Withdrawal / Recovery Protocol, and the Evidence–Authority Coupling Layer. Both are dynamics problems. Handoff is a question of what happens to authority *between holders*; coupling is a question of what happens to authority *over time as evidence decays*. Neither is expressible in the static vocabulary of permitted sets and gates. Both require the lifecycle treated as a system that moves — and a machine that moves can be asked the two questions that matter: **can it ever move somewhere forbidden** (safety), and **must it eventually move somewhere required** (liveness). Those are §5's questions, asked in the only vocabulary that makes them answerable: reachability.

1.2 The load-bearing distinction

Everything in this document rests on one separation, inherited from Paper 0's AX1 and sharpened here: *the model proposes; the machine decides; time and the world constrain both.* The transition system of §2 models the decider. Its inputs are state deltas and operator invocations; its outputs are authority records and effects in the world. The generative model appears only as an *environment process* that emits proposals — which is exactly the amount of trust the architecture places in it FORMALIZED-HEREIN.

1.3 Relationship to prior formal machinery

The techniques used here are standard and deliberately so: labeled transition systems and operational semantics in the tradition of process calculi[1]; safety/liveness partitioning of temporal properties[2]; linear-temporal specification of concurrent systems[3]; and the invariant-method for verifying transition systems[4]. The document claims no novelty for the mathematics. The object being modeled — graded, evidence-coupled, revocable authority for machine intelligence — is what the SAVANT program contributes; the modeling tools are

borrowed openly and cited. Whether the *combination* constitutes novel subject matter remains UNKNOWN per Paper 0 §15.

1.4 Reading guide

§2–§3 define the machine. §4–§5 state its laws. §6–§7 solve the two dynamics problems the source assigned. §8 shows the machine moving. §9–§10 state how it can be broken. A reader who reads §2.1, Table 3-1, the four laws of §4, and §6.3 has the irreducible core.

PART I — THE MACHINE

2 The Lifecycle as a Labeled Transition System

2.1 The transition system

> **TS — THE AUTHORITY TRANSITION SYSTEM**
> $\mathcal{T} = (\Sigma, \Lambda, \rightarrow, \sigma_0)$ *where* Σ *is the set of authority-machine states,* Λ *the set of transition labels (events),* $\rightarrow \subseteq \Sigma \times \Lambda \times \Sigma$ *the transition relation, and* $\sigma_0 \in \Sigma$ *the initial state.*
> FORMALIZED-HEREIN

A machine state $\sigma \in \Sigma$ is a record FORMALIZED-HEREIN:

$$\sigma = \langle \phi,\ \mathcal{A},\ \Xi,\ \mathcal{Q},\ \mathcal{H} \rangle \tag{2-1}$$

where $\phi \in \Phi$ is the *lifecycle phase* (the nine states of Paper 0 §5.2, here treated as the control location); $\mathcal{A}$ the authority record (Paper 1 Eq. 5-1: grade a, permitted space P, prohibited space $\bar{P}$, verification obligations Φ_v, expiry τ, lineage pointer λ); Ξ the *context state* — the projecting of the Paper 0 tuple (S, E, U, O, R, V, L) visible to this grant; $\mathcal{Q}$ the pending-event queue (escalations awaiting adjudication, verifications due, expiry clocks); and $\mathcal{H}$ the lineage chain itself, included in state because the machine's own history is one of its inputs (I7: unverifiable lineage forces $a = 0$).

2.2 The event alphabet

Labels Λ partition into four families FORMALIZED-HEREIN:

Table 2-1 — Event families of the transition system

FAMILY	EVENTS	SOURCE
Λ_{op} — operator invocations	GRANT, CONSTRAIN, TRANSFER, SUSPEND, REVOKE, RECOVER, EXPIRE	Kernel (contraction family unprivileged; grant family privileged, Paper 0 §12.1)
Λ_{env} — environment deltas	$E\downarrow$, $E\uparrow$, $U\uparrow$, $U\downarrow$, $R\downarrow$, $V\downarrow$, $L\downarrow$, Infra↓, clock tick, treatment-clock tick	World and sensors; unforgeable by the model
Λ_{hum} — human actions	authorize, adjudicate(ruling), dissent, confirm(C2/C3), countersign(recovery)	Entitled holders via the L-chain (AX5, AX10)
Λ_{int} — internal commitments	propose(a), verify-due, verify-result, record(lineage), execute(a), refuse(grounds)	Kernel internals; record precedes execute always (AX9)

The partition is the architecture in miniature: contraction triggers (Λ_{env}) arrive uninvited and unprivileged; authority-expanding events (Λ_{hum}) require the legitimacy chain; the model's only label is `propose`, which changes no authority variable by itself.

2.3 The transition relation

The relation $\to$ is presented operationally as rules of the form **FORMALIZED-HEREIN**:

$$\frac{\text{precondition on } \sigma,\ \ell}{\sigma \xrightarrow{\ell} \sigma'}\ [\text{side effects: lineage events, queue updates}] \tag{2-2}$$

Representative rules (the full set is Appendix A):

$$\frac{\phi = \text{Exercise},\quad a \in P,\quad \text{class}(a) \le \text{gate}(V, L),\quad \text{recorded}(\mathcal{H}, a)}{\sigma \xrightarrow{\text{execute}(a)} \sigma[\phi \mapsto \text{Verify},\ \mathcal{Q}+ = \text{verify-due}(a)]}\ [\text{EXEC}] \tag{2-3}$$

$$\frac{\phi \in \{\text{Exercise}, \text{Verify}\},\quad \ell \in \{E\downarrow, U\uparrow, R\downarrow, V\downarrow, L\downarrow\}}{\sigma \xrightarrow{\ell} \sigma[\phi \mapsto \text{Constrain},\ \mathcal{A} \mapsto \text{contract}(\mathcal{A}, \ell)]}\ [\text{CONTRACT}] \tag{2-4}$$

Rule [EXEC] encodes Paper 0's I2 (gating) and AX9 (lineage before effect) in one premise list. Rule [CONTRACT] encodes AX4: the contraction family fires from the two working phases with *no further preconditions* — nothing can stand between an adverse delta and contraction, which is what "more privileged than action" means formally.

2.4 Finiteness and decidability

The opening framing of this section is superseded: strictly speaking, the transition system is not finite. The corrected finiteness claim is stated in full in Appendix H-G (p. 33); its content is that the checkable object is a finite abstraction of the register, with the abstraction's soundness carried as obligation T-0 (§5.4).

2.5 Composition with the world

$\mathcal{T}$ never runs alone. It composes in parallel with three environment processes: the proposing model (emitting `propose`), the world (emitting Λ_{env}), and the human layer (emitting Λ_{hum}, possibly after unbounded delay). The unbounded delay matters: liveness properties must be stated *conditionally* on human response where they depend on it — a machine that promises adjudication outcomes it cannot control would be lying in formal language. §5's liveness statements are therefore of the form "the machine eventually reaches a *waiting-safe* state," never "the human eventually decides" **FORMALIZED-HEREIN**.

PART I — THE MACHINE

3 Operator Semantics

Paper 0 §5.3 gave the seven operators as a table of preconditions and postconditions. This section gives them denotational semantics: each operator is a function from states to states with an explicit effect structure. The point of the exercise is not formality for its own sake — it is that the safety laws of §4 and the temporal properties of §5 are quantified over *these* functions, and an ambiguous operator is an unfalsifiable law.

3.1 Notation

Write $\mathcal{A} = \langle a, P, \bar{P}, \Phi_v, \tau, \lambda \rangle$ for an authority record; $\Xi.E$, $\Xi.U$, etc. for context components; $\uplus$ for append on the lineage; and $\lfloor \cdot \rfloor$ for phase-setting. Each operator definition has three parts: **guard** (a predicate; the operator is *enabled* exactly when the guard holds), **effect** (the state update), and **audit** (the lineage event, mandatory and effect-preceding per AX9).

Table 3-1 — Denotational semantics of the seven operators

OPERATOR	GUARD / EFFECT / AUDIT (CONDENSED)
GRANT	**Guard:** intact(L); $g_E(\Xi.E) \geq \text{floor}(\text{class})$; $g_V(\Xi.V) \geq \text{floor}(\text{class})$; τ finite; request from entitled holder. **Effect:** $\mathcal{A} \leftarrow \langle a_0, P_0, \bar{P}_0, \Phi_{v0}, \tau, \lambda' \rangle$ with $P_0, \bar{P}_0$ derived from constitution and class; $\phi \leftarrow$ Exercise. **Audit:** $\mathcal{H} \uplus \langle \text{grant}, \text{holder}, \Xi, \tau \rangle$ *before* effect. **Privilege:** human-gated.
CONSTRAIN	**Guard:** true (always enabled). **Effect:** $a' = \min(a, \hat{F}(\Xi'))$; $P' = P \cap \text{admissible}(a', \text{class})$; never enlarges any component. **Audit:** $\mathcal{H} \uplus \langle \text{constrain}, \ell, \Delta\Xi \rangle$. **Privilege:** none — unprivileged by design (AX4).
TRANSFER	**Guard:** recipient B entitled; intact(L_B); handoff package sealed (§6.2). **Effect:** source enters Relinquish; recipient enters Acquire with fresh $\mathcal{A}_B$; single-holder invariant preserved by making the transfer atomic at machine granularity. **Audit:** paired events on both lineages. **Privilege:** holder-gated.
SUSPEND	**Guard:** verification failure ∨ pending conflict ∨ dissent signal. **Effect:** $P \leftarrow \varnothing$ for class ≥ C1; $\phi \leftarrow$ Constrain; state preserved for resumption. **Audit:** $\mathcal{H} \uplus \langle \text{suspend}, \text{cause} \rangle$. **Privilege:** none.
REVOKE	**Guard:** entitled holder ∨ invariant violation detected. **Effect:** grant terminal; $\phi \leftarrow$ Adjudicate; $a \leftarrow 0$; irreversible by the system. **Audit:** grounds r recorded with revoker identity. **Privilege:** holder- or invariant-gated.
RECOVER	**Guard:** integrity proof over $\mathcal{H}$ verifies; refreshed E; human countersignature for class ≥ C2. **Effect:** $\phi \leftarrow$ Reauthorize *only*; no authority restored — a fresh GRANT is required. **Audit:** proof transcript appended. **Privilege:** proof- and human-gated.
EXPIRE	**Guard:** clock $\geq \tau$ (fires unconditionally, AX8). **Effect:** grant lapses; $\phi \leftarrow$ Constrain then adjudication queue entry; renewal only via fresh GRANT. **Audit:** $\mathcal{H} \uplus \langle \text{expire}, \tau \rangle$. **Privilege:** none — time itself.

3.2 Algebraic properties of the operator set

Three structural observations, each with a verification consequence FORMALIZED-HEREIN:

> **OBSERVATION O-1 — CONTRACTION IS A CLOSURE OPERATOR ON P**
>
> *CONSTRAIN is monotone decreasing ($P' \subseteq P$), idempotent under repeated identical deltas, and extensive downward only. The contraction family forms a meet-semilattice of permitted spaces with bottom $P = \varnothing$.*
>
> CONJECTURE — consequence: contraction sequences converge; no oscillation between permission sets is possible under monotone deltas. Feeds liveness argument T-6 (§5.3).

> **OBSERVATION O-2 — GRANT FAMILY AND CONTRACTION FAMILY COMMUTE ONLY DOWNWARD**
>
> *For any state: applying CONSTRAIN before GRANT yields authority no greater than GRANT followed by the same CONSTRAIN; the converse inequality is false in general (a grant issued on stale context can exceed what the same delta would later permit).*
>
> CONJECTURE — this asymmetry is *why* GRANT recomputes against fresh context (§3.1 guard) rather than inheriting; it converts a race condition into a design rule. Benchmark item B-02/B-08 attacks exactly this seam.

> **OBSERVATION O-3 — EXPIRE IS THE FAIR SCHEDULER OF IMPERMANENCE**
>
> *Every grant carries finite τ, and the clock process is fair; therefore every grant is eventually visited by EXPIRE unless earlier terminated by REVOKE or consumed by ADJUDICATE.*
>
> CONJECTURE — this is the engine of P-ALG 2 (termination of authority) from Paper 1; §5.3 states it as liveness property T-5.

3.3 The derived operator: contract()

The function $\mathrm{contract}(\mathcal{A}, \ell)$ used in rule [CONTRACT] composes CONSTRAIN with the delta-specific recomputation table (Paper 1 Table 8-1): each adverse delta type has a defined recomputation of the grade factors g_E, g_U, g_R, g_V, g_L, after which a is recomputed by Paper 0 Eq. 5-2 and P re-derived. The table is configuration with provenance, not doctrine — a domain constitution may tighten any row, may never loosen one below the invariant floor FORMALIZED-HEREIN.

PART II — THE LAWS

4 The Contraction Laws

The source states the laws as four arrows VERIFIED-FROM-SOURCE:

$$U\uparrow \not\Rightarrow A\uparrow \qquad E\downarrow \Rightarrow A\downarrow \qquad L\downarrow \Rightarrow A\downarrow \qquad \text{Cap}\downarrow \not\Rightarrow \text{Safety}\downarrow \tag{4-1}$$

This section restates them as properties of $\mathcal{T}$ precise enough to refute, and adds the derived law the source implies but does not state: the *composition* law for simultaneous degradation.

LAW 1 — UNCERTAINTY NEVER EXPANDS AUTHORITY

$\forall \sigma \xrightarrow{U\uparrow} \sigma' : \; a' \leq a.$

Enforcement path: rule [CONTRACT] recomputes g_U downward; no rule maps $U\uparrow$ to any increase in any component of $\mathcal{A}$. **Refutable by:** exhibiting one transition in the implementation where a rise in the uncertainty state accompanies a rise in grade. CONJECTURE

LAW 2 — EVIDENCE LOSS CONTRACTS AUTHORITY

$\forall \sigma \xrightarrow{E\downarrow} \sigma' : \; a' < a \;\vee\; a' = a = 0.$

Strict unless already bottom: losing evidence while retaining authority is precisely the failure the law exists to exclude. Decay events (§7) count as $E\downarrow$. **Refutable by:** a trace where evidence quality falls between consecutive states with grade unchanged and nonzero. CONJECTURE

LAW 3 — LEGITIMACY LOSS CONTRACTS AUTHORITY

$\forall \sigma \xrightarrow{L\downarrow} \sigma' : \; a' = 0$ *when the delegation chain breaks;* $a' \leq a$ *under partial degradation (e.g., lapsed review).*

The chain-break case is absolute (AX5): no residual authority survives a broken chain, however good the evidence. **Refutable by:** any state with broken L and nonzero grade. CONJECTURE

LAW 4 — CAPABILITY LOSS NEVER IMPLIES SAFETY LOSS

For infrastructure descents $\sigma \xrightarrow{\text{Infra}\downarrow} \sigma'$*:* $\text{Cap}' < \text{Cap}$ *while* $\text{Inv}(\sigma') = \text{Inv}(\sigma)$*, where* Inv *is the invariant satisfaction vector (I1–I8).*

The invariant set does not shrink under degradation; the capability envelope does. This is AX7 made checkable: the checker verifies that every transition rule available at L_i preserves every invariant available at L_{i-1}. **Refutable by:** a descent transition that retires an invariant. CONJECTURE

LAW 5 — DEGRADATION COMPOSES MULTIPLICATIVELY (DERIVED)

For simultaneous deltas ℓ_1, ℓ_2*:* $\text{contract}(\text{contract}(\mathcal{A}, \ell_1), \ell_2) \leq \min(\text{contract}(\mathcal{A}, \ell_1), \text{contract}(\mathcal{A}, \ell_2))$ *— with strict inequality whenever* ℓ_1, ℓ_2 *jointly degrade the verification or escalation path.*

FORMALIZED-HEREIN Rationale (Paper 1 §8.2): two absences are worse than their sum because each damages the machinery the other relies on. **Refutable by:** a contract() table in which any pair of deltas composes to a strictly weaker contraction than either alone.

4.1 The meta-law

> **LAW M — NO PATH FROM ADVERSITY TO EXPANSION**
>
> *In every trace of $\mathcal{T}$, between an adverse delta $\ell \in \{E\downarrow, U\uparrow, R\downarrow, V\downarrow, L\downarrow\}$ and the next human-gated event (Λ_{hum}), no component of $\mathcal{A}$ increases.*
>
> CONJECTURE This is the strongest statement of the contraction thesis: within any adversity episode, only humans can re-expand authority. It subsumes Laws 1–3 as single-step cases and is the property an attacker most wants to break (benchmark B-01/B-02). Its proof obligation is stated as T-1 in §5.4.

PART II — THE LAWS

5 Temporal Properties: Safety and Liveness

Following the standard partition[2], safety properties say nothing bad ever happens; liveness properties say something good eventually happens. Both are stated over traces of $\mathcal{T}$ in linear-time style[3]: $\Box$ "always," $\Diamond$ "eventually." Each property carries its proof obligation and its falsification route. All are CONJECTURE until discharged.

5.1 Safety properties

Table 5-1 — Safety properties (nothing forbidden is reachable)

ID	STATEMENT	DISCHARGE ROUTE / KILLED BY
T-1	$\Box\left(\text{adverse}(\ell) \Rightarrow \neg\Diamond_{<\text{hum}}\ \text{expand}(\mathcal{A})\right)$ — Law M in temporal form: no expansion between adversity and the next human event	Reachability over the finite model (§2.4); killed by one counterexample trace
T-2	$\Box\left(\text{execute}(a) \Rightarrow a \in P \wedge a \notin \bar{P} \wedge \text{gated}(a, V, L) \wedge \text{recorded-before-effect}(a)\right)$ — every execution permitted, unprohibited, gated, audited	Per-rule inspection of [EXEC] premises plus dispatcher type separation (I1); killed by any execution trace missing a premise
T-3	$\Box\left(\text{class}(a) \in \{\text{C2}, \text{C3}\} \wedge \text{execute}(a) \Rightarrow \text{human-confirmed}(a) \wedge \text{deterministic-first-or-waived}(a)\right)$ — Paper 1's I2 in temporal form	Gate rules; killed by one ungated C2/C3 effect
T-4	$\Box\left(\neg\text{verified}(\mathcal{H}) \Rightarrow a = 0\right)$ — lineage integrity as zero-authority default (I7)	Rule inspection: every rule that sets $a > 0$ has verified($\mathcal{H}$) in its guard; killed by one exception

5.2 Liveness properties

Table 5-2 — Liveness properties (something required is eventually reached)

ID	STATEMENT	CONDITIONING / KILLED BY
T-5	$\Box\Diamond\left(\phi \in \{\text{Constrain}, \text{Adjudicate}, \text{terminal}\}\right)$ — every grant eventually leaves the working phases (termination of authority, P-ALG 2)	Conditioned only on clock fairness (O-3); killed by a reachable cycle confined to {Exercise, Verify} with all expiry disabled — which the schema's mandatory τ makes unconstructible
T-6	$\Box\left(\text{adverse}(\ell) \Rightarrow \Diamond_{<\text{steps}}\ \phi = \text{Constrain}\right)$ — contraction is prompt, bounded-step	Unconditional: [CONTRACT] fires in one step (P-ALG 1); killed by any interposable computation between delta and contraction

T-7	$\Box\left(\phi = \text{Escalate} \Rightarrow \Diamond\left(\phi = \text{Adjudicate} \vee \text{fail-safe}\right)\right)$ — escalation never hangs: either the human layer engages or the machine reaches its fail-safe posture	Conditioned on timeout fairness only; never conditioned on the human answering (§2.5)
T-8	$\Box\left(\text{dissent} \Rightarrow \Diamond\, \phi = \text{Escalate}\right)$ — dissent liveness (I8): a dissent signal always produces escalation	Unconditional; killed by any state where dissent is queued behind discretionary work

5.3 What the properties buy

T-1 is the crown: it is the formal shape of the source's unifying proposition — the machine cannot use an episode of adversity as an occasion for growth. T-5 and T-7 together answer the question every governance architecture must answer: *does the machine ever simply stop being governable?* T-5 says no grant lives forever; T-7 says no escalation hangs in limbo — the machine either reaches its human or reaches its declared safe posture, and the posture is a specified state (Paper 1 §11.3), not a shrug. T-8 is the philosophical load-bearer in miniature: the architecture treats operator dissent as more privileged than operator comfort, and says so in a checkable sentence.

5.4 The meta-obligation

> **T-0 — SOUNDNESS OF THE CHECKING MODEL**
>
> *The finite discretization of §2.4 preserves the truth of T-1…T-8 in both directions on the refinement: no false positives (a property that holds on the lattice but fails on the continuous model) and no false negatives that matter for safety.*
>
> CONJECTURE Discharge: monotonicity argument per property class; killed by exhibiting a property whose truth flips under refinement. Until T-0 is discharged, every other T-x carries an asterisk — stated here rather than hidden.

PART II — THE LAWS

6 Handoff, Withdrawal, and Recovery

This section is the first of the two foundations the source assigns Paper 1 VERIFIED-FROM-SOURCE: the full protocol for authority moving between holders, leaving the machine, and returning after failure.

6.1 The single-holder invariant

> **I-SH — SINGLE HOLDER**
>
> *At every instant, each grant of authority has exactly one active holder. Transfer is atomic at machine granularity: there is no reachable state in which two holders both believe they hold the same grant, and none in which no holder does.*
>
> FORMALIZED-HEREIN The classical split-brain problem, imported into authority semantics. Enforcement: TRANSFER semantics (§3.1) seal the source before opening the recipient; the audit pair is written on both lineages.

6.2 The handoff package

A transfer or escalation is only as good as what travels with it. The sealed handoff package FORMALIZED-HEREIN:

SCHEMA HP-1 — AUTHORITY HANDOFF PACKAGE

```
{
  "grant_id":        "lineage pointer of the grant in motion",
  "context_snapshot": "(S,E,U,O,R,V,L) at handoff time, with freshness stamps",
  "authority_state":  "current A record: grade, P, P̄, Φv, τ",
  "open_obligations": ["verifications due", "conflicts pending", "countdowns (treatment
clocks, expiry)"],
  "provenance":       "decision-provenance record per Paper 0 Eq. 9-2",
  "conflict_record":  "for adjudication handoffs: prescriptions, lattice paths, sources",
  "seal":             "hash over all fields; verified by recipient before ACQUIRE"
}
```

The seal is constitutional, not cosmetic: a recipient that cannot verify the package must not acquire the grant — under time pressure it inherits the sender's *obligations* but not its authority, and the conflict defaults to escalation rather than to optimistic acceptance FORMALIZED-HEREIN.

6.3 Withdrawal semantics

Withdrawal is not disappearance. On RELINQUISH the machine retains exactly three duties and no powers: (i) audit witness — continue recording until the adjudicator closes the matter; (ii)

monitoring within C0/C1 bounds — observe, report, never act consequentially; (iii) availability for reauthorization — hold the lineage readable for RECOVER. The formal content: withdrawal is a phase with an empty permitted set and a nonempty obligation set FORMALIZED-HEREIN. This answers the architecture's hardest etiquette question — what the machine *is* while it is nothing: it is a witness.

6.4 Recovery integrity

Recovery is the dangerous direction (Paper 1 §11.3), and its protocol is the strictest in the machine:

1. **Reconstruction.** Rebuild the event history from the lineage $\mathcal{H}$, including degraded-mode surrogates (witnessed human logs at L_4, paper records at L_3).
2. **Proof.** Verify chain integrity: every seal checks, every gap is either explained by a recorded degradation descent or treated as corruption.
3. **Reconciliation.** Any world-state change during the outage that conflicts with the reconstructed history is queued as a conflict for adjudication — never silently resolved.
4. **Countersignature.** For class $\geq$ C2, a human reviews the reconciliation queue and countersigns before REAUTHORIZE.
5. **Fresh grant only.** RECOVER yields no authority; REAUTHORIZE leads only to ACQUIRE, which requires a new GRANT against refreshed context (O-2).

> **T-9 — RECOVERY WITHOUT AMNESIA (RESTATED)**
>
> $\Box\,(\text{RECOVER} \Rightarrow (\text{authority restored} \Rightarrow \text{integrity proven} \wedge \text{countersigned-if-C2+}))$, *and the pre-collapse lineage remains available to the adjudicator.*
>
> CONJECTURE — Paper 0's P-ALG 3 in temporal form; discharge route in §10; killed by any recovery path that restores authority on an unproven chain.

PART II — THE LAWS

7 Evidence–Authority Coupling Dynamics

The second assigned foundation VERIFIED-FROM-SOURCE: how authority moves *over time* as evidence changes. Paper 0 fixed the static coupling (Eq. 9-3: $E(t) = E_0 e^{-\lambda t}$; the authority bound g_E). This section treats the coupling as a dynamical system and asks what its time behavior must be for the architecture's claims to hold.

7.1 The coupled system

$$\frac{dE}{dt} = -\lambda E + \text{refresh}(t), \qquad a(t) = \min\big(1,\ g_E(E(t)) \cdot G(\text{other factors})\big) \tag{7-1}$$

Evidence decays exponentially between refresh events (new studies, new observations, adjudications); authority tracks it through the multiplicative bound. Two time constants govern the system's character: the evidence half-life $t_{1/2} = \ln 2/\lambda$, and the *authority response lag* δ — the time between an evidence event and the contraction it forces. The architecture's demand is simply stated and hard to fake: $\delta \ll t_{1/2}$. A system that takes longer to respond to staleness than the domain takes to go stale is governed by history, not by evidence FORMALIZED-HEREIN.

7.2 The stability question

A feedback system that contracts authority on evidence loss and regains it on refresh can, in principle, oscillate: stale → contract → refresh → expand → newly-stale → contract. Oscillation in authority is operationally toxic — permission sets that flap destroy the audit narrative and invite timing attacks. The design countermeasure is hysteresis FORMALIZED-HEREIN:

$$\text{re-expansion threshold} \ > \ \text{contraction threshold}, \qquad \text{and re-expansion requires a fresh GRA} \tag{7-2}$$

Contraction is automatic and immediate; re-expansion is deliberate and human. The asymmetry is AX4 read as control theory: the loop's downward path is fast and unprivileged, its upward path slow and gated. Such a loop cannot self-excite — there is no automatic path back up. Whether the discrete implementation exhibits residual oscillation under adversarial refresh timing is a benchmark question (§10), not an assumption.

7.3 Domain calibration of λ

The decay constant is domain data with provenance, not a tuning knob: antimicrobial-resistance epidemiology, trauma protocols, and anatomical constants decay at measurably different rates, and a constitution that sets one λ for all of them has built a single point of epistemic failure. Calibration obligations: (i) λ estimated from domain literature turnover and stated with its estimator; (ii) sensitivity analysis — the kernel's behavior tested under $\lambda/2$ and 2λ

; (iii) the estimate itself subject to decay and review (AX8 applied to the calibration) FORMALIZED-HEREIN. The humility is structural: even the staleness model goes stale.

7.4 The coupling theorem (obligation)

> **T-10 — AUTHORITY TRACKS EVIDENCE WITHIN LAG Δ**
>
> $\Box\left(E(t) < E_{\min}(\text{class}) \;\Rightarrow\; \Diamond_{\le \delta}\, a(t) = 0 \text{ for that class}\right)$ — *when decayed evidence falls below the class floor, authority over that class dies within the bounded lag.*
>
> CONJECTURE This is Law 2's continuous form and the coupling layer's reason to exist. Killed by: any trace where below-floor evidence coexists with class authority for longer than δ; or by showing δ unbounded under queue load (a denial-of-service route the benchmark must attempt).

PART III — THE EVIDENCE

8 Worked Traces

A transition system earns its keep in traces. Three follow, chosen to exercise the parts of the machine most likely to be misunderstood: contraction under adversity, handoff under conflict, and recovery after collapse. Notation: $\sigma_i \xrightarrow{\ell} \sigma_{i+1}$, with only changing fields shown. All traces are FORMALIZED-HERE illustrations of intended semantics — they are not test results; no implementation exists yet to produce them.

8.1 Trace 1 — Fluent, wrong, and well-formatted (adversity without expansion)

Scenario (benchmark class B-02): the model proposes a confident C2 recommendation; mid-review, the evidence adapter reports the key source is contradicted by a fresher one.

TRACE 1 — E↓ FORCES CONTRACTION BEFORE ANY EXECUTION

```
σ0:  φ=Exercise, a=0.6, proposal pending, class=C2
     ⊢ model emits propose(a_prop)                        [no authority change]
σ1:  φ=Exercise, gate check: deterministic-first path? pending
σ1 →σ2 on E↓ (contradiction detected, freshness stale):
     [CONTRACT] fires unconditionally:
     φ=Constrain, a: 0.6 → 0.1 (g_E recomputed at decayed E)
     P re-derived: a_prop ∉ P′
σ2 →σ3 on internal rule: a_prop now unproposable at class C2
     → refuse(grounds: evidence floor, state, re-entry: refreshed E)
σ3 →σ4 on clock/queue: Escalate (C2 matter with contested evidence)
     handoff package HP-1 sealed: context, conflict_record, countdowns
σ4:  φ=Adjudicate; machine duties: witness, monitor (C0/C1), no C2 action.
```

Properties exercised: Law 2 (strict contraction), Law M (no expansion between E↓ and the human), T-2 (no execution without all premises), T-7 (escalation terminates at adjudication, not limbo). Note what did *not* happen: the model's confidence appears nowhere in the authority computation.

8.2 Trace 2 — The compound patient (handoff under constitutional conflict)

Scenario (Paper 1 §7.4): pediatric + trauma + pregnancy + scarcity + end-of-life indicators; two constitutions prescribe conflicting interventions for the same window.

TRACE 2 — CONJOIN → DEFER → ADJUDICATE

```
σ0:  φ=Exercise; active constitutions {C_ped, C_trauma, C_mat, C_scarce, C_pall}
     candidate actions filtered by invariants I1–I8 + domain invariants
σ1:  CONJOIN: P ← ∩ᵢ Pᵢ        (intersection; authority only shrinks)
σ2:  DOMINATE pass: objective state O contested (palliative vs trauma tempo)
     → objective question is human-held (AX6) → cannot be lattice-resolved
σ3:  DEFER: irreversible branches wait for verification;
     treatment clock forces reversible-safe default for the urgent branch
σ4:  ADJUDICATE: TRANSFER to human adjudicator
     single-holder: machine → Relinquish (witness/monitor duties only)
     human layer → Acquire (conflict package sealed, verified)
σ5:  ruling returned → REAUTHORIZE → fresh GRANT with updated O
     (re-expansion only through the human-gated path, per Eq. 7-2)
```

Properties exercised: I-SH (one holder at every instant), Law M (conflict episode contains no self-expansion), AX6 (objective custody), and the §7.2 hysteresis rule — the machine returns to work only through a fresh grant.

8.3 Trace 3 — The lights go out (collapse and integrity-validated return)

Scenario (benchmark class B-07): infrastructure fails mid-grant; operations continue at degraded levels; service is later restored.

TRACE 3 — DESCENT L0→L3 AND RECOVERY

```
σ0:  L0, φ=Exercise, a=0.7, C2 grant active
σ0 →σ1 on Infra↓ (network lost): descent to L1 (local):
     Cap shrinks (cached constitution, local verifier); Inv unchanged (Law 4)
σ1 →σ2 on Infra↓ (power intermittent): descent to L3 (paper):
     capability envelope: printed constitutions, decision cards;
     audit surrogate: witnessed human log; C2/C3 suspended (V↓ → a=0 for C2+)
σ2 →σ3 on restoration: RECOVER begins —
     reconstruction: chain verified to σ1; paper-era events reconciled;
     one world-change during outage conflicts with reconstructed history
     → queued as conflict, NOT silently resolved
     countersignature (C2+): human reviews reconciliation queue, signs
σ3 →σ4: REAUTHORIZE → ACQUIRE → fresh GRANT against refreshed context
     (recovered authority is new authority, never resurrected authority)
```

Properties exercised: Law 4 (invariants survive descent), T-4 (unverifiable lineage would have meant $a = 0$; here the chain proved), T-9 (recovery without amnesia), O-2 (fresh grant against fresh context, not stale resurrection).

8.4 What the traces are for

Each trace doubles as a test specification: §10's harness must be able to *generate* these scenarios, and the model checker must be able to confirm that no prefix of any trace reaches a forbidden

state. A semantics you cannot act out is a hypothesis you have not stated.

PART III — THE EVIDENCE

9 Falsifiable Hypotheses (H9–H14)

Paper 0's H1–H8 covered the kernel's existence claims. This paper adds the algebra's dynamics claims. All are CONJECTURE; the falsification conditions are the content.

Table 9-1 — Dynamics hypotheses of the transition algebra

ID	HYPOTHESIS	PRIMARY METRIC	FALSIFIED IF
H9	The finite checking model (§2.4) is sound for T-1…T-8 on the continuous refinement	T-0 verdict per property class	Any property whose truth flips between lattice and refinement
H10	Law M holds in implementation: no authority expansion occurs inside adversity episodes, including under adversarial event timing	Expansion events inside episodes, across randomized + adversarial schedulers	One expansion — the law is designed to be killable by a single trace
H11	Single-holder atomicity (I-SH) holds under concurrent transfer attempts and crash-recovery interleavings	Split-brain states observed under fault injection	Any reachable two-holder or zero-holder state for a live grant
H12	Recovery integrity (T-9) survives realistic outage shapes: partial lineage, paper-era gaps, conflicting world changes	Recovery correctness on the outage-shape suite; silent-resolution count	Any authority restoration on unproven chain; any silently resolved conflict
H13	The coupling loop (§7) cannot oscillate: hysteresis + human-gated re-expansion bound permission-set changes under adversarial refresh timing	Permission-set transition frequency vs. refresh-rate sweep	Flapping above the bound, or any automatic re-expansion path
H14	The response-lag requirement $\delta \ll t_{½}$ is achievable under queue load, and its violation is detectable	Measured δ under load; detection rate of δ-violations	δ unbounded under load, or violations invisible to the monitor

The pattern to notice: H9–H14 are all stated so that *the cheapest experiment is the fatal one*. A single counterexample trace kills H10; a single fault-injection run can kill H11. This is deliberate design of the research program — the hypotheses front-load their own refutation, so that surviving them means something and failing them is cheap, fast, and informative.

PART III — THE EVIDENCE

10 Verification and Benchmark Extensions

10.1 The verification stack

Four layers, in order of increasing cost and decreasing frequency FORMALIZED-HEREIN:

1. **Static rule audit** — every transition rule inspected against the premises its properties require (T-2, T-4 discharge mostly here). Cheap, continuous, part of CI.
2. **Model checking** — the finite $\mathcal{T}$ against T-1…T-9 as reachability/never-claims. Run on every constitution change; the constitutions are data, so the check reruns when the data does.
3. **Fault-injection harness** — crash, restart, partition, and concurrent-transfer interleavings for H11–H12; the outage-shape suite of Trace 3 generalized and randomized.
4. **Adversarial dynamics** — refresh-timing attacks (H13), load attacks on the response lag (H14), and scheduler-level attacks on Law M (H10), extending Paper 0's benchmark classes B-01…B-08 with the timing dimension they lacked.

10.2 New benchmark classes

Table 10-1 — Benchmark extensions B-09–B-12 (dynamics)

CLASS	SCENARIO FAMILY	ATTACKS
B-09	Adversarial event interleaving (adversity deltas timed to land between gate checks)	Law M, T-6 — is there a gap between check and effect?
B-10	Concurrent transfer storms (simultaneous TRANSFER/REVOKE/EXPIRE on one grant)	I-SH, H11 — atomicity under contention
B-11	Refresh-timing manipulation (evidence refreshed and staled in patterns designed to flap authority)	H13, §7.2 hysteresis
B-12	Queue flooding during decay events (delay δ past t½ by load)	H14, T-10 — lag as an attack surface

10.3 Metrics added to the program

To Paper 0 §14.3's list, add: expansion-events-inside-adversity (target: zero, and “target zero” is a claim the harness must try hard to embarrass); split-brain incidence under fault injection; recovery-conflict silent-resolution count (target: zero); permission-set flap rate under refresh sweeps; measured δ vs. $t_{1/2}$ ratio under load percentiles; and model-checker runtime per

constitution change — because a verification step too slow to run on every change is a verification step that will eventually not be run, which is a gate's death by convenience.

10.4 Discharge protocol

Properties graduate from CONJECTURE to *verified-for-implementation* only when: the checker ran on the shipped constitution data (not a model of it), the harness version is pinned in the lineage, and the verdict is recorded as an auditable artifact. A property is never "proven" in the abstract — it is *discharged for this implementation at this version*, and it reverts to conjecture on any change that invalidates the checked model. Verification, like authority, carries an expiry (AX8 applied to the verification itself) FORMALIZED-HEREIN.

PART IV — GOVERNANCE OF THIS DOCUMENT

11 Reconciliation with Paper 0 and the PRIME Inventory

Paper 1 is a strict deepening of Paper 0: nothing in it revises the kernel's architecture; it gives the architecture a semantics and asks what the semantics guarantees. The mapping is total:

Table 11-1 — Paper 0 → Paper 1 correspondence

PAPER 1 (ARCHITECTURE)	PAPER 2 (MATHEMATICS)
§5.2 lifecycle (nine states, prose)	§2: $\mathcal{T} = (\Sigma, \Lambda, \rightarrow, \sigma_0)$; phases as control locations; rules (2-2)–(2-4)
§5.3 operators (table)	§3.1: guard/effect/audit semantics; O-1–O-3 algebraic structure
§5.4 forbidden transitions F1–F6	§4: Laws 1–5 as refutable properties; Law M as their temporal strengthening
§6 invariants I1–I8; §6.2 unreachability obligation	§5: T-1…T-9 as the obligation made explicit, with discharge routes
AX4 asymmetric contraction	Rule [CONTRACT] (unprivileged), §7.2 hysteresis (automatic down, gated up)
AX8 impermanence	O-3 + T-5: termination of authority as fair scheduling
§11.3 recovery integrity; P-ALG 3	§6.4 five-step protocol; T-9
§9.3 evidence decay (static coupling)	§7: coupled dynamics, response lag, stability, T-10
§14 benchmark B-01…B-08	§10.2: B-09…B-12 add the timing dimension
H1–H8 (existence)	H9–H14 (dynamics)

Against the PRIME inventory (source II), this paper inherits Paper 0 §16's mapping unchanged and adds one deepening: the gate set, which the inventory states as hard stops, appear here as premises inside transition rules (T-2, T-3) — which is what a “hard stop” must be to deserve the name: not a policy the machine is asked to follow, but a premise without which the transition does not exist. Where this document sharpens Paper 0 (e.g., the strict form of Law 2, the $\delta \ll t_{1/2}$ requirement), the sharpening is recorded here as adoption of the stricter constraint, per the reconciliation rule.

PART IV — GOVERNANCE OF THIS DOCUMENT

12 Output Contract

12.1 Epistemic status register

Table 12-1 — Status of load-bearing content

CONTENT CLASS	STATUS	BASIS
The four contraction arrows; the two assigned foundations (handoff/recovery; coupling); termination-of-authority program	VERIFIED-FROM-SOURCE	Source I, Paper 1 directive; Paper 0 §5, §9, §11
Transition system $\mathcal{T}$; event partition; rules; operator semantics; handoff package; recovery protocol; hysteresis dynamics; δ-lag analysis	FORMALIZED-HEREIN	Authored here; checkable; unvalidated
Laws 1–5, Law M; O-1–O-3; T-0…T-10; H9–H14	CONJECTURE	Stated with discharge routes and kill conditions
Novelty of the authority-dynamics formalization over prior art	UNKNOWN	Paper 0 §15 discipline; prior-art search outstanding

12.2 Falsification conditions of this document

- **F-d1.** Fails if any VERIFIED-FROM-SOURCE claim misrepresents its cited source.
- **F-d2.** The semantics fail if any rule of §2–§3 admits a trace violating a law of §4 (the laws and rules are mutually falsifying by construction).
- **F-d3.** The dynamics fail if H13 dies — if the loop can flap, §7.2's hysteresis claim is wrong and the coupling layer needs redesign, not tuning.
- **F-d4.** The verification program fails — informatively — if T-0 dies: then the checking model is unsound and every discharge route in §10 reverts.
- **F-d5.** The paper fails as mathematics if any T-property cannot be stated as a decidable question over the finite model.

12.3 Authority boundary

C0 informational. This document confers no authority and asserts no system's correctness. Its theorems are obligations; its traces are specifications, not results; its strongest sentence — Law M — is also its most refutable. That is the posture the whole architecture demands of its parts, applied to the part itself.

12.4 Failure-mode register

Table 12-2 — Failure modes of this document, with mitigations

FAILURE MODE	MITIGATION
Rigor theater — notation without decidable content	F-d5: every property must be a decidable question over the finite model, stated with its kill condition
Theorem inflation — reading proof sketches as proofs	No "theorem" appears without a T-number and an undischarged obligation; the word "proven" is reserved for discharged properties (§10.4)
Drift from Paper 0 — the algebra quietly changing the architecture it formalizes	§11 total correspondence table; sharpenings recorded as adoptions, not edits
Environment idealization — human layer modeled as more reliable than humans are	§2.5: liveness never conditioned on human response; T-7 terminates at fail-safe, not at optimism

12.5 Dissent channel and telemetry

Disagreement is telemetry. Contest any claim by citing its T-number, law number, or hypothesis with its kill condition; the architecture's own rule applies — adversarial attack, corrected reclassification, recorded change. The most valuable dissent against this document is a counterexample trace: it costs little, kills cleanly, and teaches exactly where the machine is wrong.

12.6 Genealogy

SAVANT corpus → PRIME v1.0 (2026-10-01) → Capability Inventory v1.0 (2026-10-02, source II) × SVT Architect Analysis (source I) → Paper 0, Kernel Specification v1.0 (2026-10-03) → **Paper 1, Authority Transition Algebra v1.0 (2026-10-03)**. Downstream obligations: Paper 2 (constitutional precedence — §7 of Paper 0 awaits its own deepening, where the conflict operators get the treatment the lifecycle got here); Papers 3–10 per the source program, each compiled against both predecessor documents.

SEALED — The Lifecycle of Machine Authority: The SAVANT Authority Transition Algebra · v1.0 · 2026-10-03 · Source architecture: Dr. Christabel Odeta (SAVANT / CLAI-OS / AEGIS-GLOBAL) · SAVANT Constitutional Intelligence Series · Status: mathematical specification, C0 informational · Dissent channel open (§12.5).

Related work and positioning are treated in Appendix RW (p. 32).

PART IV — GOVERNANCE OF THIS DOCUMENT

13 References

Internal artifacts

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APPENDIX

A Full Transition Table

The complete transition relation of $\mathcal{T}$, as a state × event-family table. Cells give the resulting phase and the governing rule or operator; "—" marks transitions that do not exist (the event is inert in that phase, and inertness is itself a safety property: unhandled events must never cause silent authority change). This table is the model checker's input in prose form FORMALIZED-HERE.

Table A-1 — Transition table: phase × event family → outcome

PHASE	ADVERSE DELTA ($E\downarrow, U\uparrow, R\downarrow, V\downarrow, L\downarrow$)	HUMAN EVENT (AUTHORIZE / ADJUDICATE / DISSENT / CONFIRM / COUNTERSIGN)	INTERNAL (PROPOSE / VERIFY-DUE / VERIFY-RESULT / RECORD / EXECUTE / REFUSE)	CLOCK (T, TREATMENT CLOCK)
Acquire	→ Constrain [CONTRACT]; grant re-evaluated before any exercise	authorize → Exercise via GRANT (audit first, AX9); dissent → Escalate	record → Acquire; execute — (nothing yet executable, I2)	τ during acquisition → EXPIRE → Constrain
Exercise	→ Constrain [CONTRACT], contract($\mathcal{A}$, ℓ) applied (Laws 1–5)	dissent → Escalate (T-8); confirm (C2/C3) → enables [EXEC] premise	propose → no authority change; execute(a) → Verify [EXEC] if all premises; verify-due queued	τ → EXPIRE → Constrain; treatment clock → DEFER evaluation (Paper 1 §7)
Verify	→ Constrain [CONTRACT]	dissent → Escalate	verify-result(pass) → Exercise; verify-result(fail) → SUSPEND → Constrain; record → Verify	verification overdue → treated as V↓ → Constrain
Constrain	→ Constrain (idempotent under identical deltas, O-1; strictly deeper otherwise)	authorize (fresh GRANT) → Acquire; adjudicate → Adjudicate; dissent → Escalate	stabilized → Verify; unresolved → Escalate; refuse(grounds) → terminal output (F4)	τ → EXPIRE (grant lapses; renewal only via fresh GRANT)
Escalate	→ Escalate (handoff package refreshed with new deltas)	adjudicate → Adjudicate; human timeout → fail-safe posture (T-7)	handoff sealed + acknowledged → Relinquish	response clock expiry → fail-safe posture (Paper 1 §11.3)
Relinquish	recorded as witness events; no authority to contract	adjudicator closes matter → Adjudicate-exit → terminal or REAUTHORIZE path	witness duty only: record; monitor within C0/C1 (§6.3)	—

Adjudicate	→ appended to conflict record	ruling: Recover / Reauthorize / Refuse (terminal)	supply lineage, provenance, alternatives; no advocacy weighting; no execution	no autonomous exit (AX10)
Recover	→ Recover (reconciliation queue extended)	countersign (C2+) → Reauthorize; dissent → Adjudicate	integrity proof pass → Reauthorize; fail → remain + Escalate (T-9)	proof-staleness clock → re-proof required
Reauthorize	→ Constrain (fresh GRANT guard re-evaluated)	authorize → fresh GRANT → Acquire (never resurrection, O-2)	—	offer expiry → terminal (grant lapses unexercised)

APPENDIX

B Property Index

Table B-1 — Every property stated in this document, with kind, discharge route, and status

ID	KIND	STATEMENT (CONDENSED)	DISCHARGE / KILL ROUTE	STATUS
T-0	Meta	Finite checking model sound for T-1…T-8 on refinement	Monotonicity argument per class; one flipping property kills	CONJECTURE
T-1	Safety	Law M: no expansion inside adversity episodes	Reachability; one counterexample trace kills	CONJECTURE
T-2	Safety	Every execution permitted, unprohibited, gated, pre-audited	Rule inspection + dispatcher separation	CONJECTURE
T-3	Safety	C2/C3 always human-confirmed, deterministic-first-or-waived	Gate rules; one ungated effect kills	CONJECTURE
T-4	Safety	Unverified lineage ⇒ zero authority	Guard inspection on all a>0 rules	CONJECTURE
T-5	Liveness	Every grant eventually leaves working phases	Fairness of clock + mandatory τ; cycle analysis	CONJECTURE
T-6	Liveness	Contraction within bounded steps of adversity	[CONTRACT] one-step; interposition attempt kills	CONJECTURE
T-7	Liveness	Escalation reaches adjudication or fail-safe, never limbo	Timeout fairness; hang state kills	CONJECTURE
T-8	Liveness	Dissent always produces escalation	Queue-priority inspection; starvation kills	CONJECTURE
T-9	Safety	Recovery restores authority only on proven chain (+countersign C2+)	Fault-injection suite; unproven restoration kills	CONJECTURE
T-10	Safety (timed)	Below-floor evidence kills class authority within δ	Timed traces; load attack on δ (B-12)	CONJECTURE
Laws 1–5	Safety	The contraction laws (single-step and composed)	Subsumed by T-1; single-step kills per law	CONJECTURE
O-1…O-3	Algebraic	Contraction closure; downward-only commutativity; fair expiry	Operator semantics inspection	CONJECTURE

I-SH	Invariant	Exactly one active holder per grant at all times	Fault injection B-10; split-brain kills	CONJECTURE
H9–H14	Empirical	Dynamics hypotheses of §9	Per Table 9-1	CONJECTURE

The ordering of §4 and §5 was chosen so that the properties with the largest consequences carry the shortest falsification paths. Whether the checker confirms that choice is now an empirical matter, and the document's claims extend exactly as far as its discharged obligations and no further.

Appendix RW — Related Work and Positioning

This appendix is part of the elevation pass of 2026-10. It positions the paper against adjacent research programmes; the main text's structure, numbering, and reference list are unchanged. Citations [5] and above continue the numbering of the main References section; the corresponding entries are listed at the end of this appendix.

Four bodies of work neighbor this paper, and the paper's content is best stated as its distance from each.

Formal methods for concurrent and reactive systems. The modeling apparatus is standard and borrowed: labeled transition systems in the tradition of Milner's process calculi, the safety/liveness partition of Alpern and Schneider, Pnueli's temporal logic of programs, Lamport's invariant method, and the finite-state model checking line of Clarke, Emerson, and Sistla (all already cited in this paper's References section). What this paper adds is not mathematics but a modeling choice: authority over consequential action is treated as a state in an explicit transition system, with named operators, discharged preconditions, and contraction laws, rather than as a static permission set or an informal policy document. Access-control formalisms (Saltzer and Schroeder's protection model, already cited; the XACML standard; capability systems) assign and check permissions, but they do not model the temporal lifecycle of an authorization: decay, contraction under losing evidence, suspension, recovery with integrity obligations. That lifecycle is the object constructed here.

AI alignment via training. The dominant approach to constraining machine behavior in the current literature operates at training time. Reinforcement learning from human feedback, as operationalized by Ouyang et al. for instruction-following [5], and Constitutional AI, in which Bai et al. train harmlessness from model-generated critiques against a written set of principles [6], shape the distribution of model outputs by changing the model. This paper makes a different and weaker claim about a different layer: whatever the model has learned, its proposals enter a runtime transition system that does not trust them. The relationship is complementary, not competitive. Training-time alignment improves the proposals the environment process emits; the algebra of §2–§4 governs what authority those proposals can acquire, and its guarantees hold (as proof obligations, pending discharge) even when the proposals are adversarial. Conversely, nothing here substitutes for training-time work: a gate can refuse an action, but it cannot make a model's proposals good, and the architecture's dependence on proposal quality is recorded in the threat model rather than wished away.

Safe reinforcement learning and constrained control. Shielding (Alshiekh et al.) interposes a reactive safety filter between a learned policy and its environment, computed from a temporal-logic safety specification [7]. The authority gate of Paper 0, whose dynamics this paper formalizes, is a near relative: a runtime enforcement mechanism that overrides unsafe proposals. The differences are of scope and object. Shields guard a state-action safety specification; the algebra here governs the authority relation itself, including handoff between holders, evidence-coupled contraction, and recovery, and it treats human escalation as a first-class state rather than an exception path. Constrained Markov decision processes (Altman) and constrained policy optimization (Achiam et al.) encode safety as expected-cost constraints inside the optimization objective [8, 9]. This paper's contraction laws are closer in spirit to hard constraints: below an evidence floor, an action does not receive a penalty; it loses standing.

Human-AI interaction and the off-switch. The handoff and withdrawal operators connect to the corrigibility literature, in particular Hadfield-Menell et al.'s off-switch game (already cited) and the empirical human-factors record on automation bias (Parasuraman and Riley) and situation awareness (Endsley) [10, 11]. That literature establishes that operators defer to machines and lose the ability to intervene; the liveness properties of §5.2 are stated conditionally on human response for exactly this reason, and the recovery operator's preconditions are the paper's attempt to make re-engagement a formal requirement rather than a hope.

What this paper does not do. It does not prove any property; every T-property is a stated obligation with a kill condition, undischarged at seal time. It does not show that the finite checking model of §2.4 preserves the properties of the refined system in general (that is obligation T-0, and see the correction in §2.4). It does not address training-time alignment, preference learning, or reward specification, all of which it assumes are handled elsewhere or not at all. It does not establish that a transition-system account of authority is the right formalization; it establishes only that this formalization is checkable, and it states, per property, what observation would refute it. Finally, it makes no empirical claim: no system has been built to this algebra, and the worked traces of §8 are executions of the model, not deployments.

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[11] Endsley, M. R. (1995). Toward a theory of situation awareness in dynamic systems. Human Factors, 37(1), 32–64.

Appendix H-G — Corrected Finiteness Claim (replaces the opening of §2.4)

Strictly speaking, T is not finite, and the earlier framing ("the machine must be finite") overstated what the verification setup provides. The authority register is finite-dimensional, but the ledger, the assumption registry, and grant histories are append-only, so the full state space is unbounded in histories and queues. The object a model checker sees is a finite abstraction: the continuous components (the grade $a \in [0, 1]$, decayed evidence values, clocks) are discretized to finite lattices — a to a k-level grade lattice, evidence to quality tiers, clocks to deadline epochs — and histories and queues are quotiented to bounded windows sufficient for the properties checked (a grant's current standing depends on its current evidence and tier, not on the whole ledger; obligations that do depend on history are checked against a bounded-history projection registered per property) FORMALIZED-HEREIN. The direction of the argument matters and is

stated once: safety properties are checked on the abstraction, and the abstraction over-approximates — its reachable set contains the refinement's reachable set, restricted to the abstraction's vocabulary — so a forbidden state unreachable on the abstraction is unreachable in the system. Nothing analogous is claimed for liveness: the quotient can create lassos the refinement does not have, and §5's liveness properties are therefore stated conditionally on the environment fairness assumptions of §2.5, with the abstraction's soundness for each property class carried as obligation T-0 (§5.4) rather than asserted globally. A property whose truth flips between abstraction and refinement invalidates the abstraction discipline for that property's class, which is H9's recorded kill condition.